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Bilayer seal material for air gaps in semiconductor devices

Abstract

The present disclosure relates to a method for forming a semiconductor device includes forming an opening between first and second sidewalls of respective first and second terminals. The first and second sidewalls oppose each other. The method further includes depositing a first dielectric material at a first deposition rate on top portions of the opening and depositing a second dielectric material at a second deposition rate on the first dielectric material and on the first and second sidewalls. The second dielectric material and the first and second sidewalls entrap a pocket of air. The method also includes performing a treatment process on the second dielectric material.

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Background/Summary

(1) This application is a continuation of U.S. Non-Provisional Application Ser. No. 16/937,237, titled “Bilayer Seal Material for Air Gaps in Semiconductor Devices,” which was filed on Jul. 23, 2020 which claims the benefit of U.S. Provisional Patent Application No. 62/951,809, titled “Bilayer Seal Material for Air Gaps in Semiconductor Devices,” which was filed on Dec. 20, 2019, which are incorporated herein by reference in their entireties.

BACKGROUND

(1) The semiconductor integrated circuit (IC) industry has experienced exponential growth. Technological advances in IC materials and design have produced generations of ICs, where each generation has smaller and more complex circuits than the previous generation. In the course of the IC evolution, functional density (e.g., the number of interconnected devices per chip area) has generally increased while geometry size (e.g., the smallest component or line that can be created using a fabrication process) has decreased. This scaling process provides benefits by increasing production efficiency and lowering associated costs.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the common practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of illustration and discussion.

(2) FIG. 1 is an isometric view of a semiconductor structure, in accordance with some embodiments.

(3) FIGS. 2-6 are cross-sectional views of various partially formed semiconductor structures, in accordance with some embodiments.

(4) FIG. 7 is a flow diagram of a method of forming bilayer seal structures in semiconductor structures, in accordance with some embodiments.

(5) FIG. 8 illustrates chemical reactions occurring during the formation of bilayer seal structures in semiconductor structures, in accordance with some embodiments.

DETAILED DESCRIPTION

(6) The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely

examples and are not intended to be limiting. For example, the formation of a first feature over a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features are disposed between the first and second features, such that the first and second features are not in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

(7) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper,” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

(8) The term “nominal” as used herein refers to a desired, or target, value of a characteristic or parameter for a component or a process operation, set during the design phase of a product or a process, together with a range of values above and/or below the desired value. The range of values is typically due to slight variations in manufacturing processes or tolerances.

(9) The terms “about” and “substantially” as used herein indicate the value of a given quantity that can vary based on a particular technology node associated with the subject semiconductor device. In some embodiments, based on the particular technology node, the terms “about” and “substantially” can indicate a value of a given quantity that varies within, for example, 5% of a target value (e.g., $\pm 1\%$, $\pm 2\%$, $\pm 3\%$, $\pm 4\%$, $\pm 5\%$ of the target value).

(10) The fins may be patterned by any suitable method. For example, the fins may be patterned using one or more photolithography processes, including double-patterning or multi-patterning processes. Generally, double-patterning or multi-patterning processes combine photolithography and self-aligned processes, allowing patterns to be created that have, for example, pitches smaller than what is otherwise obtainable using a single, direct photolithography process. For example, in one embodiment, a sacrificial layer is formed over a substrate and patterned using a photolithography process. Spacers are formed alongside the patterned sacrificial layer using a self-aligned process. The sacrificial layer is then removed, and the remaining spacers may then be used to pattern the fins.

(11) As planar semiconductor devices, such as metal-oxide-semiconductor field effect transistors (“MOSFETs”), are scaled down through various technology nodes, other approaches to increase device density and speed have been advanced. One approach is the fin field effect transistor (“finFET”) device that is a three-dimensional FET that includes the formation of a fin-like channel extending from the substrate. FinFETs are compatible with conventional complementary metal-oxide-semiconductor (CMOS) processes and their three-dimensional structure allows them to be aggressively scaled while maintaining gate control and mitigating short channel effects. Gate stacks are used in planar and three-dimensional FETs for controlling the conductivity of the semiconductor device. A gate stack including gate dielectric layer and gate electrode for a finFET device can be formed by a replacement gate process where a polysilicon sacrificial gate structure is replaced by a metal gate structure. Gate dielectric layers, such as a high-k dielectric layer (e.g., a dielectric layer having dielectric constant greater than about 3.9), is formed between the channel and the gate electrode. Spacers can be disposed on sidewalls of the gate stack to protect the gate structures during fabrication processes, such as ion implantation, gate replacement process, epitaxial source/drain structure formation, and other suitable processes. Air gaps can be used in place of spacers to reduce the effective dielectric constant that in turn can reduce parasitic capacitance and improve device performance. Air gaps can be formed by depositing a seal material over an opening between terminals of a semiconductor device such that a pocket of air is trapped

between the terminals. As the dielectric constant of air is generally lower than a dielectric material, the effective dielectric constant can be reduced. However, seams in the seal material can lead to defects in the semiconductor device. For example, fabrication processes for forming air gap structures often involve multiple etching and cleaning processes that can etch through portions of the seal material through the seams and cause damage to the air gaps, such as causing the collapse of the seal material or trapping chemical solutions within the air gap. The damaged air gap structure can cause defects in the semiconductor device and lead to low device yield and even device failure.

(12) To address the above shortcomings, the present disclosure provides a semiconductor device and method of fabricating the same to provide simple and cost-effective structures and process for producing seamless seal layers in semiconductor devices. The seamless seal layers can be used to seal an opening and form air gaps between terminals of semiconductor devices. Specifically, a bilayer seal material can be formed by depositing a first seal material, depositing a second seal material, and performing at least one treatment process on the deposited first and second seal materials. The first and second seal materials can be dielectric materials. In some embodiments, the first and second seal materials can be formed using silicon oxycarbide (SiCO). The first seal material is deposited on portions of opposing sidewalls towards the top of an opening and a second seal material is deposited on the first seal material and on exposed surfaces in the opening. The second seal material is deposited on the first seal materials that are on the opposing sidewalls. The deposition process of the second seal material lasts at least until the second seal material from opposing sidewalls are merged to form an enclosed space between the opposing sidewalls. A treatment process can be performed on the deposited first and second seal materials such that seams are removed by the expansion of at least the second seal material. In some embodiments, the treatment process can be an anneal process performed in an oxygen ambient environment. In some embodiments, the first seal material can be deposited at a greater deposition rate than that of the second seal material. In some embodiments, the first and second seal materials can be formed using precursors, such as tetramethyldisiloxane (TMDSO), hydrogen, oxygen, and any other suitable precursors. Forming seal materials by depositing a bilayer seal material, such as silicon oxycarbide, followed by a treatment process on the deposited bilayer seal material can prevent damaging underlying structures, such as the oxidation of metal source/drain structures.

(13) FIG. 1 is an isometric view of exemplary fin field effect transistors (finFETs) structures. FIGS. 2-7 provide various exemplary semiconductor structures and fabrication processes that illustrate the formation of multi-spacer structures having air gaps and seal materials, in accordance with some embodiments. The fabrication processes provided herein are exemplary, and alternative processes in accordance with this disclosure can be performed (though they are not shown in these figures).

(14) FIG. 1 is an isometric view of a finFET, according to some embodiments. FinFET **100** can be included in a microprocessor, memory cell, or other integrated circuit. The view of finFET **100** in FIG. 1 is shown for illustration purposes and may not be drawn to scale. FinFET **100** may include further suitable structures, such as additional spacers, liner layers, contact structures, and any other suitable structures, are not illustrated in FIG. 1 for the sake of clarity.

(15) FinFET **100** can be formed on a substrate **102** and can include a fin structure **104** having fin regions **121** and S/D regions **106**, gate structures **108** disposed on fin structures **104**, spacers **110** disposed on opposite sides of each of gate structures **108**, and shallow trench isolation (STI) regions **112**. FIG. 1 shows five gate structures **108**. However, based on the disclosure herein, finFET **100** can have more or fewer gate structures. In addition, finFET **100** can be incorporated into an integrated circuit through the use of other structural components—such as S/D contact structures, gate contact structures, conductive vias, conductive lines, dielectric layers, and passivation layers—that are omitted for the sake of clarity.

(16) Substrate **102** can be a semiconductor material, such as silicon. In some embodiments, substrate **102** includes a crystalline silicon substrate (e.g., wafer). In some embodiments, substrate

102 includes (i) an elementary semiconductor, such as germanium; (ii) a compound semiconductor including silicon carbide, gallium arsenide, gallium phosphide, indium phosphide, indium arsenide, and/or indium antimonide; (iii) an alloy semiconductor including silicon germanium carbide, silicon germanium, gallium arsenic phosphide, gallium indium phosphide, gallium indium arsenide, gallium indium arsenic phosphide, aluminum indium arsenide, and/or aluminum gallium arsenide; or (iv) a combination thereof. Further, substrate **102** can be doped depending on design requirements (e.g., p-type substrate or n-type substrate). In some embodiments, substrate **102** can be doped with p-type dopants (e.g., boron, indium, aluminum, or gallium) or n-type dopants (e.g., phosphorus or arsenic).

(17) Fin structure **104** represents current-carrying structures of finFET **100** and can traverse along a Y-axis and through gate structures **108**. Fin structure **104** can include: (i) portions of fin regions **121** underlying gate structures **108**; and (ii) S/D regions **106** disposed on portions of fin regions **121** that are formed on opposing sides of each of gate structures **108**. Portions of fin regions **121** of fin structure **104** under gate structures **108** (not shown in FIG. 1) can extend above STI regions **112** and can be wrapped around by corresponding one of gate structures **108**. Fin regions **121** on opposing sides of gate structures **108** can be etched back such that S/D regions **106** can be epitaxially grown on the etched back portions of fin regions **121**.

(18) Fin regions **121** of fin structure **104** can include material similar to substrate **102**. S/D regions **106** can include an epitaxially-grown semiconductor material. In some embodiments, the epitaxially-grown semiconductor material is the same material as substrate **102**. In some embodiments, the epitaxially-grown semiconductor material includes a different material from substrate **102**. The epitaxially-grown semiconductor material can include: (i) a semiconductor material, such as germanium and silicon; (ii) a compound semiconductor material, such as gallium arsenide and aluminum gallium arsenide; or (iii) a semiconductor alloy, such as silicon germanium and gallium arsenide phosphide. Other materials for fin structure **104** are within the scope of this disclosure.

(19) In some embodiments, S/D regions **106** can be grown by (i) chemical vapor deposition (CVD), such as by low pressure CVD (LPCVD), ultrahigh vacuum CVD (UHVCVD), reduced pressure CVD (RPCVD), or a suitable CVD process; (ii) molecular beam epitaxy (MBE) processes; (iii) a suitable epitaxial process; and (iv) combinations thereof. In some embodiments, S/D regions **106** can be grown by an epitaxial deposition/partial etch process, which repeats the epitaxial deposition/partial etch process at least once. Such repeated deposition/partial etch process is also called a “cyclic deposition-etch (CDE) process.” In some embodiments, S/D regions **106** can be grown by selective epitaxial growth (SEG), where an etching gas is added to promote the selective growth of semiconductor material on the exposed surfaces of fin structures, but not on insulating material (e.g., dielectric material of STI regions **112**). Other methods for epitaxially growing S/D regions **106** are within the scope of this disclosure.

(20) S/D regions **106** can be p-type regions or n-type regions. In some embodiments, p-type S/D regions **106** can include SiGe and can be in-situ doped during epitaxial growth using p-type dopants, such as boron, indium, and gallium. For p-type in-situ doping, p-type doping precursors, such as diborane (B.sub.2H.sub.6), boron trifluoride (BF.sub.3), and other p-type doping precursors, can be used. In some embodiments, n-type S/D regions **106** can include Si and can be in-situ doped during an epitaxial growth process using n-type dopants, such as phosphorus and arsenic. For n-type in-situ doping, n-type doping precursors, such as phosphine (PH.sub.3), arsine (AsH.sub.3), and other n-type doping precursors, can be used. In some embodiments, S/D regions **106** are not in-situ doped, and an ion implantation process is performed to dope S/D regions **106**.

(21) Spacer **110** can include spacer portions **110a** that form on sidewalls of gate structure **108** and are in contact with dielectric layer **118**, spacer portions **110b** that form on sidewalls of fin structure **104**, and spacer portions **110c** that form as protective layers on STI regions **106**. Each spacer portion can also be a multi-spacer structure including more than one spacer structure. For example,

spacer portion **110a** can include more than one spacer and an air gap formed between gate structure **108** and fin structure **104**. A seal material can be formed over the air gap to enclose and protect the air gap from subsequent fabrication processes. The air gap and seal material are not shown in FIG. **1** for simplicity. Spacers **110** can include insulating material, such as silicon oxide, silicon nitride, a low-k material, and a combination thereof. Spacers **110** can have a low-k material with a dielectric constant less than 3.9 (e.g., less than 3.5, 3, and 2.8). As air gaps can have dielectric constant about 1, the effective dielectric constant of spacers **110** can be further reduced compared to spacers formed using low-k material. The low-k material for spacers **110** can be formed using suitable deposition processes, such as an atomic layer deposition (ALD). In some embodiments, spacers **110** can be deposited using CVD, LPCVD, UHVCVD, RPCVD, physical vapor deposition (PVD), any other suitable deposition processes, and combinations thereof. The seal material can be formed by depositing a first seal material on top portions of an opening formed between gate structures **108** and S/D regions **106**, followed by a deposition of second seal material on the first seal material to form an enclosure having air trapped in the opening. Other materials and thicknesses for spacers **110** and seal material are within the scope of this disclosure.

(22) Each gate structure **108** can include a gate electrode **116**, a dielectric layer **118** adjacent to and in contact with gate electrode **116**, and a gate capping layer **120**. Gate structures **108** can be formed by a gate replacement process.

(23) In some embodiments, dielectric layer **118** can be formed using a high-k dielectric material (e.g., dielectric material having dielectric constant greater than about 3.9). Dielectric layer **118** can be formed by CVD, atomic layer deposition (ALD), physical vapor deposition (PVD), e-beam evaporation, or other suitable processes. In some embodiments, dielectric layer **118** can include (i) a layer of silicon oxide, silicon nitride, and/or silicon oxynitride, (ii) a high-k dielectric material, such as hafnium oxide (HfO₂), TiO₂, HfZrO, Ta₂O₃, HfSiO₄, ZrO₂, and ZrSiO₂, (iii) a high-k dielectric material having oxides of lithium (Li), beryllium (Be), magnesium (Mg), calcium (Ca), strontium (Sr), scandium (Sc), yttrium (Y), zirconium (Zr), aluminum (Al), lanthanum (La), cerium (Ce), praseodymium (Pr), neodymium (Nd), samarium (Sm), europium (Eu), gadolinium (Gd), terbium (Tb), dysprosium (Dy), holmium (Ho), erbium (Er), thulium (Tm), ytterbium (Yb), or lutetium (Lu), or (iv) a combination thereof. High-k dielectric layers can be formed by ALD and/or other suitable methods. In some embodiments, dielectric layer **118** can include a single layer or a stack of insulating material layers. Other materials and formation methods for dielectric layer **118** are within the scope of this disclosure. For example, portions of dielectric layer **118** are formed on horizontal surfaces, such as top surface of STI regions **112**. Although not visible in FIG. **1**, dielectric layer **118** can also be formed on top and sidewalls of fin regions **121** that are under gate electrode **116**. In some embodiments, dielectric layer **118** is also formed between sidewalls of gate electrode **116** and spacer portions **110a**, as shown in FIG. **1**. In some embodiments, dielectric layer **118** have a thickness **118t** in a range of about 1 nm to about 5 nm.

(24) Gate electrode **116** can include a gate work function metal layer **122** and a gate metal fill layer **124**. In some embodiments, gate work function metal layer **122** is disposed on dielectric layer **118**. Gate work function metal layer **122** can include a single metal layer or a stack of metal layers. The stack of metal layers can include metals having work functions similar to or different from each other. In some embodiments, gate work function metal layer **122** can include, for example, aluminum (Al), copper (Cu), tungsten (W), titanium (Ti), tantalum (Ta), titanium nitride (TiN), tantalum nitride (Ta₂N₃), nickel silicide (NiSi), cobalt silicide (CoSi), silver (Ag), tantalum carbide (TaC), tantalum silicon nitride (TaSiN), tantalum carbon nitride (TaCN), titanium aluminum (TiAl), titanium aluminum nitride (TiAlN), tungsten nitride (WN), metal alloys, and combinations thereof. Gate work function metal layer **122** can be formed using a suitable process, such as ALD, CVD, PVD, plating, or combinations thereof. In some embodiments, gate work function metal layer **122** has a thickness **122t** in a range from about 2 nm to about 15 nm. Other materials, formation

methods, and thicknesses for gate work function metal layer **122** are within the scope of this disclosure.

(25) Gate metal fill layer **124** can include a single metal layer or a stack of metal layers. The stack of metal layers can include metals different from each other. In some embodiments, gate metal fill layer **124** can include a suitable conductive material, such as Ti, Ag, Al, TiAlN, TaC, TaCN, TaSiN, Mn, Zr, TiN, TaN, Ru, Mo, WN, Cu, W, Co, Ni, TiC, TiAlC, TaAlC, metal alloys, and combinations thereof. Gate metal fill layer **124** can be formed by ALD, PVD, CVD, or other suitable deposition processes. Other materials and formation methods for gate metal fill layer **124** are within the scope of this disclosure.

(26) In some embodiments, gate capping layer **120** can have a thickness **120t** in a range from about 5 nm to about 50 nm and can protect gate structure **108** during subsequent processing of finFET **100**. Gate capping layer **120** can include nitride material, such as silicon nitride, silicon-rich nitride, and silicon oxynitride. Other materials for gate capping layer **120** are within the scope of this disclosure.

(27) STI regions **112** can provide electrical isolation to finFET **100** from neighboring active and passive elements (not illustrated herein) integrated with or deposited onto substrate **102**. STI regions **112** can have a dielectric material, such as silicon oxide, silicon nitride, silicon oxynitride, fluorine-doped silicate glass (FSG), a low-k dielectric material, and other suitable insulating materials. In some embodiments, STI regions **112** can include a multi-layered structure. The cross-sectional shapes of fin structure **104**, S/D regions **106**, gate structures **108**, spacers **110**, and STI regions **112** are illustrative and are not intended to be limiting.

(28) FIGS. 2-6 provide various exemplary semiconductor structures and fabrication processes that illustrate the formation of spacer structures having air gaps and seamless seal layers, in accordance with some embodiments. FIG. 7 is a flow diagram of a method **700** of forming air gaps and seamless seal layers in semiconductor structures, in accordance with some embodiments of the present disclosure. Based on the disclosure herein, other operations in method **700** can be performed. Further, the operations of method **700** can be performed in a different order and/or vary.

(29) The air gaps with seamless seal layers can provide the benefit of reducing and/or eliminating damage to the air gaps formed between spacer structures. The fabrication processes can be used to form planar semiconductor devices or vertical semiconductor devices, such as finFETs. In some embodiments, the fabrication processes illustrated in FIGS. 2-7 can be used to form semiconductor structures similar to finFET structures described above in FIG. 1. For example, the semiconductor structures illustrated in FIGS. 2-7 can be similar to finFET **100** during different stages of fabrication as viewed from the cut A-A' illustrated in FIG. 1.

(30) Referring to operation **702** of FIG. 7, source/drain regions and gate stacks are formed on a substrate, according to some embodiments. FIG. 2 is a cross-sectional view of a semiconductor structure **200** after three neighboring gate structures **208** and two source/drain contacts **230** are formed over a substrate. The substrate can include fin region **221**. Each gate stack such as gate structure **208** includes a gate dielectric layer **218** and a gate electrode **216**. Gate dielectric layer **218** can be formed on sidewalls and bottom surfaces of gate electrode **216**. Channel regions for semiconductor devices, such as finFETs, can be formed in fin region **221** and under gate structures **208**.

(31) Fin region **221** can be current-carrying semiconductor structures formed on the substrate. For example, fin region **221** can be similar to fin region **121** described above in FIG. 1. In some embodiments, fin region **221** can include a semiconductor material, such as germanium, silicon, silicon carbide, gallium arsenide, gallium phosphide, indium phosphide, indium arsenide, indium antimonite, silicon germanium carbide, silicon germanium, gallium arsenic phosphide, gallium indium phosphide, gallium indium arsenide, gallium indium arsenic phosphide, aluminum indium arsenide, aluminum gallium arsenide, any suitable material, and combinations thereof. In some embodiments, fin region **221** can be doped with p-type or n-type dopants.

(32) Gate dielectric layer **218** can be formed on fin region **221** and formed using a high-k dielectric material. Gate dielectric layer **218** can be deposited by CVD, ALD, PVD, e-beam evaporation, or other suitable processes. In some embodiments, gate dielectric layer **218** can include a high-k dielectric material, such as HfO_2 . In some embodiments, gate dielectric layer **218** can include TiO_2 , HfZrO , Ta_2O_3 , HfSiO_4 , ZrO_2 , and ZrSiO_2 . In some embodiments, gate dielectric layer **218** can be similar to dielectric layer **118** described above in FIG. 1.

(33) Gate electrode **216** can be formed on gate dielectric layer **218** and can include a single metal layer or a stack of metal layers. Gate structures **208** can further include work function layers and are not illustrated in FIG. 2 for simplicity. The stack of metal layers can include metals having work functions similar to or different from each other. In some embodiments, gate electrode **216** can be formed of a conductive material, such as Al, Cu, W, Ti, Ta, TiN, TaN, NiSi, CoSi, Ag, TaC, TaSiN, TaCN, TiAl, TiAlN, WN, metal alloys, and combinations thereof. Gate electrode **216** can be formed using a suitable deposition process, such as ALD, CVD, PVD, plating, and combinations thereof. Other materials and formation methods for gate electrode **216** are within the scope of this disclosure. In some embodiments, gate electrode **216** can be formed using a gate replacement process, where a polysilicon gate is removed and a metal gate electrode is formed in the place of the removed polysilicon gate.

(34) Spacer structures can be formed on sidewalls of gate structures **208**. In some embodiments, gate structures can include a gate electrode, dielectric layers, spacers, any other suitable structures, and are collectively referred to as gate structures for ease of reference. In some embodiments, spacers **210** and **212** can be formed on sidewalls of gate dielectric layer **218** and on top surfaces of fin region **221**. Spacer structures are formed on to protect gate structure **208** during subsequent processing. In some embodiments, spacer **210** can have an L-shaped cross section with a vertical portion formed on the sidewall of gate dielectric layer **218** and a horizontal portion formed on the top surface of fin region **221**. Spacer **210** can be formed using a dielectric material, such as silicon carbide nitride, silicon nitride, silicon oxide, any suitable dielectric material, and combinations thereof. In some embodiments, the carbon atomic content can be less than about 30% for spacer **210** formed using silicon carbide nitride. In some embodiments, the carbon atomic content of spacer **210** can be between about 20% and about 30%. Additional spacers, such as spacer **212**, can also be formed. For example, spacer **212** can be formed on the horizontal portion of spacer **210**, on the top surface of fin region **221**, or both. In some embodiments, spacer **212** can be formed using a dielectric material, such as silicon. In some embodiments, the materials that form spacers **210** and **212** can have high etch selectivity (e.g., greater than about 10) such that when spacer **212** is removed spacer **210** can remain substantially intact. In some embodiments, spacers **210** and **212** can be formed using any suitable dielectric material, such as silicon nitride, silicon oxynitride, silicon carbide, silicon oxycarbide, silicon on glass (SOG), tetraethoxysilane (TEOS), PE-oxide, HARP formed oxide, and combinations thereof. In some embodiments, spacers **210** and **212** can be formed using a low-k dielectric material.

(35) Source/drain (S/D) regions **240** can be formed in fin region **221**. S/D regions **240** can be p-type regions or n-type regions. In some embodiments, p-type S/D regions **240** can include SiGe and can be in-situ doped during an epitaxial growth process using p-type dopants, such as boron, indium, and gallium. For p-type in-situ doping, p-type doping precursors, such as B_2H_6 , BF_3 , and other p-type doping precursors, can be used. In some embodiments, n-type S/D regions **240** can include Si and can be in-situ doped during an epitaxial growth process using n-type dopants, such as phosphorus and arsenic. For n-type in-situ doping, n-type doping precursors, such as PH_3 , AsH_3 , and other n-type doping precursors, can be used. In some embodiments, S/D regions **240** are not in-situ doped, and an ion implantation process is performed to dope S/D regions **240**. In some embodiments, S/D regions can be similar to S/D regions **160** described above in FIG. 1.

(36) Source/drain (S/D) contacts **230** can be in physical and electrical contact with source/drain regions **240**. S/D contacts **230** can be formed by depositing a conductive material between adjacent gate structures **208**. For example, openings can be formed between spacers **212** to expose underlying S/D regions **240**. A deposition process can be performed to deposit the conductive material in the openings such that electrical connections can be made. In some embodiments, a contact etch stop layer (CESL) **214** can be deposited in the opening prior to the deposition of the conductive materials. Examples of the conductive material deposition process can include PVD, sputtering, electroplating, electroless plating, any suitable deposition process, and combinations thereof. A planarization process can be performed after the deposition process such that top surfaces of gate electrode **216**, spacers **210** and **212**, CESL **214**, and source/drain contacts **230** can be substantially coplanar (e.g., an even surface). In some embodiments, S/D contacts **230** can be formed using tungsten, aluminum, cobalt, silver, any suitable conductive material, and combinations thereof.

(37) Similar to the finFET **100** described in FIG. 1, semiconductor structure **200** can be formed on a substrate where fin regions **221** protrude from STI regions. The STI regions are not visible from the cross-sectional view of semiconductor structure **200** illustrated in FIG. 2, but a top surface of the STI regions is represented by dashed line **222** for ease of description.

(38) Referring to operation **704** of FIG. 7, one or more spacers are removed to form openings between terminals of the semiconductor device, according to some embodiments. FIG. 3 is a cross-sectional view of the semiconductor device after one or more spacers are removed to form openings. One or more spacers of the spacers between gate electrode **216** and S/D contacts **230** can be removed. For example, spacers **212** can be removed to form openings **302** that is surrounded by spacer **210** and CESL **214**. One or more etching processes can be used to remove spacer **212**. In some embodiments, an etching process that has high etch selectivity of spacer **212** over other structures in semiconductor structure **200** can be used to remove spacer **212** while keeping the other exposed structures intact. For example, spacers **212** can be formed using silicon carbide nitride, and a wet etching process and/or a plasma etching can be used to selectively remove spacers **212**. For example, spacers **212** can be formed using silicon material and a capacitively coupled plasma (CCP) etching process using nitrogen tetrafluoride and hydrogen as precursors and be used to selectively remove spacers **212**. In some embodiments, the plasma etching process can be performed at a temperature between about 0° C. and about 100° C. For example, the processing temperature can be between about 0° C. and about 30° C., between about 30° C. and about 60° C., between about 60° C. and about 100° C., or any suitable temperature range. In some embodiments, the plasma etching process can be performed at a chamber pressure between about 500 mTorr and about 5 TOM For example, the chamber pressure can be between about 500 mTorr and about 2 Torr, between about 2 Torr and about 5 Torr, or any suitable pressure.

(39) FIGS. 4A-4C are cross-sectional views illustrating seamless seal material formed in the openings of the semiconductor device, according to some embodiments. FIGS. 4A-4C are enlarged views of region **304** of FIG. 3. Other structures can be included in the structures shown in FIGS. 4A-4C and are not illustrated for simplicity.

(40) Referring to operation **706** of FIG. 7, a first seal material is deposited on at least corners of openings in the semiconductor device, according to some embodiments. FIG. 4A is a cross-sectional view illustrating the semiconductor device after the first seal material is deposited. First seal material **412** is deposited on top surfaces of gate electrode **216**, gate dielectric layer **218**, S/D contacts **230**, and CESL **214**. In some embodiments, first seal material **412** can also be deposited in opening **302**. For example, first seal material **412** can be deposited on sidewalls of spacer **210** and CESL **214**. In some embodiments, first seal material **412** can be deposited on the bottom of opening **302**, such as on the top surface of the horizontal portion of spacer **210** formed on fin region **221**. In some embodiments, first seal material **412** can also be formed on fin region **221** if a portion of the top surface of fin region **221** is exposed between spacer **210** and CESL **214**. First seal

material **412** can include a corner portion **412A** formed on spacer **210** and CESL **214**. Top surfaces of spacer **210** and CESL **214** can respectively have rounded corners **410A** and **414A** to facilitate the growth of corner portion **412A** of first seal material **412**. The curved surfaces of rounded corners **410A** and **414A** can reduce the formation of voids or discontinuities in first seal material **412** compared to corners having right angles or sharp edges. Corner portions **412A** of first seal material **412** can contour the curved surfaces of rounded corners **410A** and **414A**. First seal material can have horizontal portions **412B** formed on the top surfaces of gate electrode **216**, gate dielectric layer **218**, and S/D contacts **230** to protect them from subsequent fabrication processes. For example, horizontal portions **412B** can prevent oxidation of underlying materials during subsequent etching or treatment processes. A thickness $T_{sub.1}$ of horizontal portions **412B** can be between about 10 nm and about 40 nm. In some embodiments, thickness $T_{sub.1}$ can be between about 10 nm and about 15 nm, between about 15 nm and about 30 nm, between about 30 nm and about 40 nm, or any suitable thicknesses. Increasing thickness $T_{sub.1}$ can provide the benefits of providing greater protection for underlying structures (e.g., metal contacts or gate structures) from subsequent processing, such as etching or cleaning processes. In addition, thickness $T_{sub.1}$ can be adjusted to provide a nominal opening profile of opposing corner portions **412A**. For example, increasing thickness $T_{sub.1}$ can reduce the distance between opposing corner portions **412A**, which in turn can form an air gap with a greater volume by reducing the amount of the second seal material deposited below corner portions **412A**.

(41) First seal material **412** can affect the volume of subsequently formed air gaps between gate electrode **216** and S/D contacts **230** by adjusting the depth of first seal material **412** that extends into opening **302**. Specifically, corner portions **412A** of first seal material **412** can extend into opening **302** by forming on sidewalls of spacers **210** and CESL **214**. Openings **302** can have depth $H_{sub.1}$, and a greater extension depth $H_{sub.2}$ of corner portions **412A** into opening **302** can provide a smaller subsequently formed air gap (not shown in FIG. 4A) in opening **302**. For example, a greater value of the ratio $H_{sub.2}$ over $H_{sub.1}$ can leave less volume in opening **302** for air gaps to be formed. In some embodiments, opening **302** has a height $H_{sub.1}$ that can be between about 30 nm and about 60 nm. In some embodiments, extension depth $H_{sub.2}$ can be between about 10 nm and about 20 nm. In some embodiments, the ratio of $H_{sub.1}$ over $H_{sub.2}$ can be between about 11 and about 2. In some embodiments, a planarization process such as a chemical mechanical polishing process (CMP) can be performed on the first seal material **412**, and heights $H_{sub.1}$ and $H_{sub.2}$ may be reduced. For example, after the planarization process, height $H_{sub.1}$ can be between about 25 nm and about 55 nm, and height $H_{sub.2}$ can be between about 5 nm and about 15 nm.

(42) First seal material **412** can be formed using any suitable dielectric material. In some embodiments, first seal material **412** can be formed using material that provides sufficient mechanical strength to support the air gap structure and chemical resistance to protect from subsequent chemical processes. In some embodiments, first seal material **412** can include silicon-oxygen or silicon-carbon cross-links. For example, first seal material **412** can be formed using a silicon oxycarbide material. The oxygen and carbon atomic contents of the silicon oxycarbide material can be adjusted to achieve various properties of first seal material **412**. For example, increasing the oxygen atomic content of first seal material **412** can reduce current leakage in first seal material **412**. Increasing the carbon atomic content in first seal material **412** can provide increased etch selectivity of first seal material **412** over adjacent structures, such as spacers **210** and **214**. In some embodiments, the silicon atomic content of first seal material **412** formed using silicon oxycarbide can be between about 28% and about 35%. For example, the silicon atomic content can be between about 28% and about 31%, between about 31% and about 35%, or any suitable range. In some embodiments, the oxygen atomic content of first seal material **412** can be between about 35% and about 52%. For example, the oxygen atomic content can be between about 35% and about 45%, between about 45% and about 52%, or any suitable range. In some

embodiments, the carbon atomic content of first seal material **412** can be between about 15% and about 35%. For example, the carbon atomic content can be between about 15% and about 25%, between about 25% and about 35%, or any suitable ranges. In some embodiments, first seal material **412** can be deposited using radical CVD, CVD, ALD, LPCVD, UHVCVD, RPCVD, PVD, any other suitable deposition processes, and combinations thereof. In some embodiments, first seal material **412** can be deposited using a radical CVD process with an ion filter.

(43) Referring to FIG. 7, the deposition of first seal material **412** can include one or more operations. For example, the deposition can include a first operation **706A** of flowing precursors into a deposition chamber. The precursors can provide one or more of the following bonding types: silicon-oxygen, silicon-hydrogen, and silicon-carbon. In some embodiments, the precursors are in gas phase and can include, for example, tetramethyldisiloxane (TMDSO) hydrogen gas, and oxygen gas. Other suitable precursors can also be included. The flow ratio of hydrogen gas over oxygen gas can be greater than about 20 to minimize the oxidation of underlying materials while facilitating the chemical reactions needed for the deposition. For example, the flow ratio of hydrogen gas over oxygen gas can be between about 20 and about 30. The deposition can further include a second operation **706B** that includes activating plasma and used to activate the precursors in their gas phase to form silicon-oxygen and silicon-carbon cross-links. The deposition process can include a third operation **706C** of a treatment process to reduce the oxygen content from the deposited seal material. The treatment process can be performed in a hydrogen chamber environment. In some embodiments, the treatment process can be performed in chamber environments having any suitable type of gas, such as argon, nitrogen, and any suitable gas. In some embodiments, the deposition process can be performed at a temperature between about 300° C. and about 700° C. For example, the deposition temperature can be between about 300° C. and about 500° C., between about 500° C. and about 700° C., and at any suitable temperature. In some embodiments, the deposition and treatment process can be performed in cycles, such as a cyclic process deposition-treatment process. For example, the deposition and treatment process can be followed by another deposition and treatment process until a nominal thickness or quality of first seal material has been achieved. In some embodiments, the cyclic process can include flowing precursors into the deposition chamber and performing a series of activation/treatment processes. In some embodiments, the cyclic process can include performing in series the following: flowing precursors, activating the precursors, and performing the treatment. In some embodiments, the flowing of precursors can be performed between a series of activation/treatment processes. FIG. 8 illustrates exemplary chemical reactions occurring during the deposition of first seal material **412**.

(44) The deposition rate can be adjusted through various deposition parameters. A greater deposition speed can facilitate greater accumulation of first seal material at curved surfaces **410A** and **414A**. A lower deposition speed can provide a greater extension depth H.sub.2 of first seal material **412** into opening **302**. A greater deposition speed can be achieved through adjusting various suitable processing parameters. In some embodiments, the deposition process can be performed at a deposition rate greater than about 25 Å/min. For example, the deposition process can be performed at a rate between about 25 Å/min and about 35 Å/min. In some embodiments, the deposition rate can be between about 55 Å/min and about 65 Å/min. For example, the deposition rate can be about 60 Å/min. In some embodiments, a lower chamber pressure during deposition or greater plasma power can provide a greater deposition rate. In some embodiments, chamber pressure can be between about 0.5 Torr and about 12 Torr. For example, chamber pressure can be between 0.5 Torr and about 3 Torr, between about 3 Torr and about 7 Torr, between about 7 Torr and about 12 Torr, and any other suitable ranges/values. As another example, a chamber pressure between about 4.5 Torr and about 5.5 Torr can provide deposition rate of about 35 Å/min while a chamber pressure between about 6 Torr and about 7 Torr can provide a lower deposition rate at about 20 Å/min.

(45) The plasma power level for the deposition can also affect the deposition rate. A greater plasma

power level can provide a greater deposition rate. In some embodiments, the plasma power level can be between about 500 W and about 3000 W. For example, the plasma power level can be between about 500 W and about 1000 W, between about 1000 W and about 2000 W, between about 2000 W and about 3000 W, and at any other suitable power levels.

(46) The density of first seal material **412** can also be adjusted through deposition parameters. Increasing the density of seal material **412** can provide for greater mechanical support and improved chemical resistance. In some embodiments, first seal material **412** can have a density greater than about 2.0 g/cm³. For example, the density of first seal material **412** can be between about 2 g/cm³ and about 2.2 g/cm³. In some embodiments, the density can be between about 2.2 g/cm³ and about 3.2 g/cm³. In some embodiments, a greater density can be achieved through lower chamber processing pressure and greater plasma power level. In some embodiments, the chamber processing pressure can be between about 0.5 Torr and about 12 Torr. For example, the chamber processing pressure can be between about 0.5 Torr and about 3 Torr, between about 3 Torr and about 8 Torr, between about 8 Torr and about 12 Torr, and any other suitable ranges or values. In some embodiments, the plasma power level can be between about 500 W and about 3000 W. For example, the plasma power level can be between about 500 W and about 2000 W, between about 2000 W and about 3000 W, and any other suitable ranges or values. In some embodiments, the deposition process can use radical triggered chemical reaction with an ion filter. Using an ion filter in a plasma deposition process can improve the conformity of the deposited first seal material **412**.

(47) The dielectric constant of first seal material **412** can be less than about 5. In some embodiments, first seal material **412** can have a dielectric constant between about 3.2 and about 5. A lower dielectric constant of first seal material **412** can lead to lower parasitic capacitance of the terminals of semiconductor device **200**. In some embodiments, the leakage current in semiconductor structure **200** can be less than about 1E⁻⁸ A/cm² at 2 MV/cm.

(48) An optional treatment process can be performed on first seal material **412** to further increase the amount of its internal crosslinks and/or improve its density. For example, a hydrogen anneal process can be performed to reduce the oxygen content and can form additional Si—C—Si bonds in first seal material **412**. The hydrogen treatment process can also remove chemical byproducts, such as H₂O. In some embodiments, the optional treatment process can be performed for less than about 1 min and greater than about 5 s. For example, the treatment process can be performed for between about 40 s and about 1 min. In some embodiments, the optional treatment process can be performed at a chamber pressure lower than about 5 Torr. For example, the chamber pressure can be set at about 3 Torr. In some embodiments, the optional treatment process can be performed at an elevated temperature. For example, the processing temperature can be set at greater than about 300° C. In some embodiments, the processing temperature can be about 350° C. In some embodiments, increasing the treatment time, reducing chamber pressure, and/or increasing processing temperature can increase the density and result in more cross links of first seal material **412**.

(49) Referring to operation **708** of FIG. 7, a second seal material is deposited on the first seal material and in the openings, according to some embodiments. FIG. 4B is a cross-sectional view illustrating the semiconductor device after the second seal material is deposited. Second seal material **432** is deposited on portions of surfaces of first seal material **412**, spacer **210**, and CESL **214**. Second seal material **432** can include at least: (i) corner portions **432A** deposited on corner portions **412A** of first seal material **412**; (ii) horizontal portion **432B** deposited on **412B** of first seal material **412**, and (iii) vertical portions **432C** deposited on sidewalls of spacer **210** and CESL **214**. In some embodiments, second seal material **432** can be deposited on the bottom of opening **302**, such as on the top surface of the horizontal portion of spacer **210** formed on fin region **221**.

(50) Second seal material **432** can be deposited using any suitable deposition process. For example, second seal material **432** can be deposited using a CVD process. Semiconductor structure **200** can

be loaded into a deposition chamber and a seal material is subsequently blanket deposited. As precursors in the deposition chamber have to move through the opening formed between opposing corner portions **412A** of first seal material **412** to be deposited on exposed surfaces of opening **302**, the precursors have lower probabilities to come into contact with surfaces of spacers **210** and CESL **214** compared to the top surfaces of horizontal portions **412B**. Accordingly, the seal material is deposited at a much lower rate in opening **302** that is below corner portions **412A**. As the seal material gradually accumulates on opposing corner portions **412A** of first seal material **412** to form corner portions **432A** of second seal material **432**, corner portion **432A** being deposited over one corner portion **412A** would merge at region **440** with another corner portion **432A** deposited over an opposing corner portion **412A**. At region **440**, a seam **450** is formed between the adjacent corner portions **432A** of second seal material **432**. A thickness $T_{sub.2}$ of horizontal portions **432B** can be between about 20 nm and about 50 nm. In some embodiments, thickness $T_{sub.2}$ can be between about 20 nm and about 30 nm, between about 30 nm and about 40 nm, between about 40 nm and about 50 nm, or any suitable thicknesses. In some embodiments, a total of thicknesses $T_{sub.1}$ and $T_{sub.2}$ can be between about 40 nm and about 80 nm. In some embodiments, the total of thicknesses $T_{sub.1}$ and $T_{sub.2}$ can be greater than 80 nm. The deposition of second seal material **432** can proceed until the opening between corner portions **412A** of first seal material **412** has been closed. Increasing thickness $T_{sub.2}$ can provide the benefits of ensuring the opening between corner portions **412A** has been closed by the deposition of second seal material **432** and an air gap has been formed.

(51) Second seal material **432** can affect the volume of subsequently formed air gaps between gate electrode **216** and S/D contacts **230** by adjusting the depth of second seal material **432** that extends into opening **302**. Specifically, vertical portions **432C** of second seal material **432** can extend into opening **302** by forming on sidewalls of spacers **210** and CESL **214**. A distance $H_{sub.3}$ between seam **450** and the bottom surface of opening **302** can be between about 20 nm and about 50 nm. A greater depth $H_{sub.3}$ can provide a greater air gap **442** formed between gate electrode **216** and S/D contacts **230**. A distance $H_{sub.4}$ between the lower end of vertical portion **432C** and the bottom surface of opening **302** can be between about 0 and about 45 nm.

(52) Second seal material **432** can be formed using any suitable dielectric material. In some embodiments, second seal material **432** can be formed using material that provides sufficient bonding strength to first seal material **412**. In some embodiments, second seal material **432** can include silicon-oxygen or silicon-carbon cross-links. For example, second seal material **432** can be formed using a silicon oxycarbide material. In some embodiments, the silicon, oxygen, and carbon atomic contents of second seal material **432** after its deposition process can be similar to those of first seal material **412**. In some embodiments, those atomic contents can be different between first and second seal materials **412** and **432**. In some embodiments, second seal material **432** can be deposited using radical CVD, CVD, ALD, LPCVD, UHVCVD, RPCVD, PVD, any other suitable deposition processes, and combinations thereof. In some embodiments, second seal material **432** can be deposited using a radical CVD process with an ion filter. In some embodiments, the deposition of second seal material **432** can be similar to the deposition process of first seal material **412**. For example, the deposition process of second seal material **432** can include a first operation **708A** by flowing precursors into a deposition chamber. In some embodiments, second seal material **432** can be formed by a CVD process using precursors that include, for example, tetramethyldisiloxane (TMDSO) hydrogen gas, and oxygen gas. Other suitable precursors can also be used. The flow ratio of hydrogen gas over oxygen gas can be greater than about 20 to minimize the oxidation of underlying materials while facilitating the chemical reactions needed for the deposition. For example, the flow ratio of hydrogen gas over oxygen gas can be between about 20 and about 30. The deposition can further include a second operation **708B** that includes activating plasma and used to activate the precursors in their gas phase to form silicon-oxygen and silicon-carbon cross-links. In some embodiments, the deposition process can be performed at a

temperature between about 300° C. and about 700° C. For example, the deposition temperature can be between about 300° C. and about 450° C., between about 450° C. and about 700° C., and at any other suitable temperatures. The deposition process can further include a third operation **708C** where a treatment process is performed on second seal material **432**. In some embodiments, the treatment process can be an anneal process performed in an oxygen environment. In some embodiments, the treatment process can be similar to the treatment process described below with reference to operation **710**. In some embodiments, the treatment processes can be different. In some embodiments, third operation **708C** can include an anneal process performed in a deposition chamber filled with non-reactant gases, such as argon. In some embodiments, third operation **708C** can be a treatment process performed using hydrogen. In some embodiments, the cyclic process can include flowing precursors into the deposition chamber and performing a series of activation/treatment processes. In some embodiments, the cyclic process can include performing in series the following: flowing precursors, activating the precursors, and performing the treatment. In some embodiments, the flowing of precursors can be performed between a series of activation/treatment processes. FIG. 8 illustrates exemplary chemical reactions occurring during the deposition of second seal material **432**.

(53) The deposition rate can be adjusted through various deposition parameters. Second seal material **432** can be deposited at a lower deposition rate than first seal material **412**. In some embodiments, second seal material **432** can be a substantially conformal film deposition over corner portions **412A** and horizontal portion **412B** of first seal material **412**. A greater deposition speed can facilitate greater accumulation of second seal material at corner portions **412A**. A lower deposition speed can provide a greater extension of second seal material **432** into opening **302**. A greater deposition speed can be achieved through adjusting various suitable processing parameters. In some embodiments, the deposition process can be performed at a deposition rate less than about 30 Å/min. For example, the deposition process can be performed at a rate between about 20 Å/min and about 30 Å/min. In some embodiments, a lower chamber pressure during deposition or greater plasma power can provide a greater deposition rate. In some embodiments, the chamber pressure can be between about 0.5 Torr and about 12 Torr. For example, chamber pressure can be between 0.5 Torr and about 3 Torr, between about 3 Torr and about 7 Torr, between about 7 Torr and about 12 Torr, and any other suitable ranges/values.

(54) The plasma power level for the deposition can also affect the deposition rate. A greater plasma power level can provide a greater deposition rate. In some embodiments, the plasma power level can be between about 500 W and about 3000 W. For example, the plasma power level can be between about 500 W and about 1000 W, between about 1000 W and about 2000 W, between about 2000 W and about 3000 W, and at any other suitable power levels.

(55) The density of second seal material **432** can also be adjusted through deposition parameters. Increasing the density of second seal material **432** can provide for greater mechanical support and improved chemical resistance. In some embodiments, second seal material **432** can have a density greater than about 2.0 g/cm.³. For example, the density of second seal material **432** can be between about 2 g/cm.³ and about 2.5 g/cm.³. In some embodiments, the density can be between about 2.2 g/cm.³ and about 2.5 g/cm.³. In some embodiments, a greater density can be achieved through lower chamber processing pressure and greater plasma power level. In some embodiments, the chamber processing pressure can be between about 0.5 Torr and about 12 Torr. For example, the chamber processing pressure can be between about 0.5 Torr and about 3 Torr, between about 3 Torr and about 8 Torr, between about 8 Torr and about 12 Torr, and any other suitable ranges or values. In some embodiments, the plasma power level can be between about 500 W and about 3000 W. For example, the plasma power level can be between about 500 W and about 2000 W, between about 2000 W and about 3000 W, and any other suitable ranges or values. In some embodiments, the deposition process can use radical triggered chemical reaction with an ion filter. Using an ion filter in a plasma deposition process can improve the conformity of the

deposited second seal material **432**.

(56) The dielectric constant of second seal material **432** can be the same or different from first seal material **412**. For example, second seal material **432** can have a dielectric constant less than about 5. In some embodiments, second seal material **432** can have a dielectric constant between about 3.2 and about 5. In some embodiments, the leakage current in semiconductor structure **200** can be less than about $1\text{E}.\text{sup}.-8\text{ A/cm}.\text{sup}.\text{2}$ at 2 MV/cm.

(57) Referring to operation **710** of FIG. 7, a treatment process is performed on the first and second seal materials of the seal layer, according to some embodiments. FIG. **4C** is a cross-sectional view illustrating the semiconductor device after the treatment process is performed. A treatment process **435** can be performed on second seal material **432** to remove seams, such as seams **450**. For example, an oxygen anneal process can be performed such that second seal material **432** physically expands and forms additional bonds at seam **450**. During the oxygen anneal process, a portion of the Si—C—Si bonds in second seal material **432** can become Si—O—Si bonds. In some embodiments, the total carbon atomic ratio of second seal material **432** can decrease between about 5% and about 15%. In some embodiments, treatment process **435** can alter the silicon, oxygen, and carbon atomic contents of second seal material **432**. For example, prior to treatment process **435**, second seal material **432** can have an atomic content composition similar to that of first seal material **412**. After treatment process **435**, the atomic content of second seal material **432** can change. For example, treatment process **435** using an oxygen anneal process can increase the oxygen atomic content in second seal material **432**. Therefore, second seal material **432** can have a higher oxygen atomic content than that of first seal material **412**. In some embodiments, increasing the oxygen atomic content in second seal material **432** can lead to the physical expansion of second seal material **432** that results in additional bonds formed at seam **450**. In some embodiments, the oxygen atomic content of second seal material **432** can be between about 40% and about 60%. For example, the oxygen atomic content can be between about 40% and about 45%, between about 45% and about 50%, between about 50% and about 60%, or any suitable ranges. In some embodiments, the carbon atomic content of second seal material **432** can be lower than that of first seal material **412**. For example, the carbon atomic content of second seal material **432** can be between about 10% and about 25%. In some embodiments, the carbon atomic content can be between about 10% and about 15%, between about 15% and about 25%, or any suitable ranges. In some embodiments, the silicon atomic content of second material **432** formed using silicon oxycarbide can be between about 25% and about 40%. For example, the silicon atomic content can be between about 25% and about 33%, between about 33% and about 40%, or any suitable ranges. The oxygen treatment process can be performed for less than about 1 min. For example, the treatment process can be performed for between about 40 s and about 1 min. In some embodiments, the oxygen flow rate for treatment process **435** can be between about 1 sccm and about 10 sccm. For example, the oxygen flow rate can be between about 1 sccm and about 3 sccm, between about 3 sccm and about 5 sccm, between about 5 sccm and about 10 sccm, and any other suitable values. The oxygen anneal process can remove any seams such as seams **450** such that region **440** contains second seal material **432** without any seams.

(58) Referring to operation **712** of FIG. 7, a planarization process is performed on the seamless seal layer, according to some embodiments. FIG. **5** is a cross-sectional view of a semiconductor device after the planarization process is performed. As shown in FIG. **5**, seamless seal material **532** is formed on semiconductor structure **200**, entrapping a pocket of air to form air gaps **542** between terminals of semiconductor structure **200** and a substrate such as fin region **221**. Seamless seal material **532** can be formed between and in physical contact with spacer **210** and CESL **214**. Seamless seal material **532** can also be in contact with other structures not illustrated in FIG. **5**. A planarization process can be used to remove portions of first and second seal materials **412** and **432**. The planarization process can continue until the top surfaces of gate electrode **216**, gate dielectric layer **218**, spacer **210**, CESL **214**, and S/D contacts **230** are exposed and are substantially

level. After the planarization process, the remaining portions of first and second seal materials **412** and **432** can form seamless seal material **532**. An air pocket entrapped by seamless seal material **532** can form air gaps **542** between terminals of semiconductor structure **200** such as gate structure **208** and S/D contacts **230**. In some embodiments, air gaps **542** can include different types of air. For example, air gaps **542** can include oxygen, hydrogen, helium, argon, nitrogen, any other suitable types of air, and combinations thereof. A lower deposition rate of seamless seal material **532** can result in air gaps **542** having smaller volumes. For example, seamless seal material **532** can be formed by depositing first seal material **412** and second seal material **432**, and a lower deposition rate of second seal material **432** can provide an air gap **542** having shorter height that results in a smaller air gap volume. As air gaps **542** can have a dielectric constant of about 1, the effective dielectric constant of spacer **210** and air gap **542** can be lower compared to a spacer structure consisting of spacers **210** and **214**.

(59) Referring to operation **714** of FIG. **7**, dielectric layers and interconnect structures are formed, according to some embodiments. FIG. **6** is a cross-sectional view illustrating dielectric layers and interconnect structures formed on the semiconductor device.

(60) A dielectric layer **620** can be formed on the top surfaces of gate electrode **216**, gate dielectric layer **218**, spacer **210**, seamless seal material **532**, CESL **214**, S/D contacts **230**, and other suitable structures. In some embodiments, dielectric layer **620** can be an etch stop layer. Dielectric layer **620** can be formed using a low-k dielectric material (e.g., dielectric layer having a dielectric constant lower than about 3.9), such as silicon oxide. An inter-layer dielectric (ILD) layer **650** can be formed on dielectric layer **620**. ILD layer **650** can be formed of a low-k dielectric material. For example, ILD layer **650** can be formed using silicon oxide. In some embodiments, dielectric layer **620** and ILD layer **650** can be formed using CVD, ALD, PVD, flowable CVD (FCVD), sputtering, any suitable deposition process, and combinations thereof. Contacts can be formed in ILD **650** to establish electrical connection from S/D contacts **230** and gate electrode **216** to external circuitry, such as peripheral circuits formed above semiconductor structure **200**. Gate vias **616** can be formed in ILD **650** and extend through dielectric layer **620** to be in physical contact with gate electrode **216**. Similarly, S/D vias **630** can extend through ILD **650** and in physical contact with S/C contacts **230**. Gate vias **616** and S/D vias **630** can be formed by a patterning and etching process. For example, openings can be formed in ILD **650** and through dielectric layer **620** to expose gate electrode **216** and S/D contact **216**, respectively. A deposition process can be performed to deposit conductive material in the openings such that electrical connections can be made. Examples of the deposition process can be PVD, sputtering, electroplating, electroless plating, any suitable deposition process, and combinations thereof. A planarization process can be performed after the deposition process such that top surfaces of ILD **650**, gate vias **616**, and S/D vias **630** can be substantially coplanar (e.g., level). In some embodiments, gate vias **616** and S/D vias **630** can be formed using tungsten, aluminum, cobalt, silver, any suitable conductive material, and combinations thereof.

(61) Various embodiments of the present disclosure provide semiconductor devices and methods of fabricating the same to provide simple and cost-effective structures and process for producing seamless seal layers in semiconductor devices. The seamless seal layers can be used to seal an opening and form air gaps between terminals of semiconductor devices to reduce effective dielectric constant that in turn can improve device performance. A bilayer seal material can be formed by depositing a first seal material, depositing a second seal material, and performing at least one treatment process on the deposited first and second materials.

(62) In some embodiments, a method for forming a semiconductor device includes forming an opening between first and second sidewalls of respective first and second terminals. The first and second sidewalls oppose each other. The method further includes depositing a first dielectric material at a first deposition rate on top portions of the opening and depositing a second dielectric material at a second deposition rate on the first dielectric material and on the first and second

sidewalls. The second dielectric material and the first and second sidewalls entrap a pocket of air.

The method also includes performing a treatment process on the second dielectric material.

(63) In some embodiments, a method for forming a semiconductor device includes forming a gate structure and a source/drain (S/D) contact on a substrate. The method also includes depositing a first dielectric material. A first portion of the first dielectric material is on a top portion of a sidewall of the gate structure. A second portion of the first dielectric material is on a top portion of a sidewall of the S/D contact. The method further includes depositing a second dielectric material. A first portion of the second dielectric material is on the first portion of the first dielectric material and on the sidewall of the gate structure. A second portion of the second dielectric material is on the second portion of the first dielectric material and on the sidewall of the S/D contact. Depositing the second dielectric material continues until the first and second portions of the second dielectric material are in contact with one another. The method further includes performing an oxygen treatment process on the deposited second dielectric material.

(64) In some embodiments, a method for forming a semiconductor device includes forming an opening over a top surface of a substrate and between first and second terminals of the semiconductor device. The method further includes depositing a first dielectric material. A first portion of the first dielectric material is on a top portion of a sidewall of the first terminal. A second portion of the first dielectric material is on a top portion of a sidewall of the second terminal. The method also includes depositing a second dielectric material. A first portion of the second dielectric material is on the first portion of the first dielectric material. A second portion of the second dielectric material is on the second portion of the first dielectric material. A pocket of air is entrapped in the opening surrounded by the second dielectric material, the first and second terminals, and the substrate. The method also includes performing an oxygen treatment process on the deposited second dielectric material.

(65) It is to be appreciated that the Detailed Description section, and not the Abstract of the Disclosure, is intended to be used to interpret the claims. The Abstract of the Disclosure section may set forth one or more but not all exemplary embodiments contemplated and thus, are not intended to be limiting to the subjoined claims.

(66) The foregoing disclosure outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art will appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art will also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the subjoined claims.

Claims

1. A semiconductor device, comprising: a gate structure and a source/drain structure on a substrate; a first sidewall structure on a sidewall of the gate structure; a second sidewall structure on a sidewall of the source/drain structure, wherein the second sidewall structure is opposite to the first sidewall structure; a first dielectric layer disposed on the first and second sidewall structures; and a second dielectric layer disposed on the first dielectric layer and on the first and second sidewall structures, wherein the second dielectric layer and the first and second sidewall structures entrap a pocket of air, and wherein top surfaces of the first sidewall structure, the second sidewall structure, the first dielectric layer, and the second dielectric layer are substantially coplanar.

2. The semiconductor device of claim 1, wherein the first sidewall structure is formed of a dielectric material.

3. The semiconductor device of claim 1, wherein the first dielectric layer and the second dielectric

layer comprise silicon-oxygen or silicon-carbon cross links.

4. The semiconductor device of claim 1, wherein a third dielectric layer is formed on top surfaces of the first sidewall structure, the second sidewall structure, the first dielectric layer, and the second dielectric layer.

5. A semiconductor device, comprising: a gate structure and a source/drain (S/D) contact on a substrate; a first sidewall structure on a sidewall of the gate structure; a second sidewall structure on a sidewall of the S/D contact; a first dielectric layer comprising a first portion on a top portion of the first sidewall structure and a second portion on a top portion of the second sidewall structure; a second dielectric layer in contact with the first portion and second portion of the first dielectric layer, wherein top surfaces of the first sidewall structure, the second sidewall structure, the first dielectric layer, and the second dielectric layer are substantially coplanar; and an air gap enclosed by the first sidewall structure, the second sidewall structure, and the second dielectric layer.

6. The semiconductor device of claim 5, wherein the second dielectric layer separates the first and second portions of the first dielectric layer.

7. The semiconductor device of claim 5, wherein: the first sidewall structure is an L-shaped spacer, and the second sidewall structure is an etch stop layer.

8. The semiconductor device of claim 7, wherein a horizontal portion of the L-shaped spacer is in contact with a bottom portion of the etch stop layer.

9. The semiconductor device of claim 8, further comprising a portion of the second dielectric layer on the horizontal portion of the L-shaped spacer.

10. The semiconductor device of claim 5, wherein the air gap comprises at least one of oxygen, nitrogen, hydrogen, helium, argon, or a combination thereof.

11. A semiconductor device, comprising: a first terminal and a second terminal; a first sidewall structure on a sidewall of the first terminal; a second sidewall structure on a sidewall of the second terminal, wherein the first and second sidewall structures define boundaries of an air gap and have different shapes; a first dielectric layer comprising a first portion on a top portion of the first sidewall structure and a second portion on a top portion of the second sidewall structure; and a second dielectric layer comprising a first portion in contact with the first dielectric layer, a second portion on a sidewall surface of the first sidewall structure, and a third portion on a sidewall surface of the second sidewall structure, wherein top surfaces of the first sidewall structure, the second sidewall structure, and the first portion of the second dielectric layer are substantially coplanar.

12. The semiconductor device of claim 11, wherein the first sidewall structure has an L-shape.

13. The semiconductor device of claim 11, wherein the second sidewall structure comprises an etch stop layer.

14. The semiconductor device of claim 11, wherein the first sidewall structure is in contact with the second sidewall structure.

15. The semiconductor device of claim 11, wherein the first sidewall structure, the second sidewall structure, and the second dielectric layer enclose the air gap.

16. The semiconductor device of claim 15, wherein the air gap comprises at least one of oxygen, nitrogen, hydrogen, helium, argon, or a combination thereof.

17. The semiconductor device of claim 10, further comprising a third dielectric layer disposed on top surfaces of the first terminal, the second terminal, the first sidewall structure, the second sidewall structure, the first dielectric layer and the second dielectric layer.

18. The semiconductor device of claim 1, wherein the first sidewall structure has an L-shape.

19. The semiconductor device of claim 1, wherein the first sidewall structure is in contact with the second sidewall structure.

20. The semiconductor device of claim 5, further comprising a third dielectric layer formed on top surfaces of the first sidewall structure, the second sidewall structure, the first dielectric layer, and the second dielectric layer.
